ON AN ALGEBRAIC STACK, THE DIRECT SUM OF SHIFTED LOCALLY QUASI-COHERENT SHEAVES WITH FLAT BASE CHANGE PROPERTY EQUALS THEIR PRODUCT IN THE DERIVED CATEGORY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The proof delegates to lemma-is-limit (same chapter, source 4990-5007) with 'we see that the direct sum is isomorphic to the product' - the bridge (truncations of the direct sum are finite partial sums; the derived-limit triangle realizes the product) is standard mechanical book-keeping, content verbatim.

Editorial note. The label lemma-sum-is-product names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Locally quasi-coherent sheaves with the flat base change property on an algebraic stack (Sheaves on Stacks / Cohomology of Stacks definitions)
- Same-chapter lemma-is-limit (source 4990-5007): an object K of D(O_X) whose cohomology sheaves are locally quasi-coherent with flat base change is the derived limit of its canonical truncations tau_{>= -n} K
- Direct sums vs products in derived categories; truncation of a direct sum of sheaves shifted in distinct degrees

Source reference. Original reference 08IY: sources/stacks-examples/pinned-original-source.tex, source lines 4990–4990. Prerequisite pointer invoked at 5037; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack. If $\mathcal{F}_n$ is a collection of locally quasi-coherent sheaves with the flat base change property on $\mathcal{X}$, then $\oplus_n \mathcal{F}_n[n] \to \prod_n \mathcal{F}_n[n]$ is an isomorphism in $D(\mathcal{O}_\mathcal{X})$.*

Proof. This is true because by Lemma 08IY we see that the direct sum is isomorphic to the product. □

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